MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 2C53

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: High-Mu triode

Outline	---	See figure 1
Base	---	B8-26
Cap	---	C1-1
Envelope	---	T9
Cathode	---	Coated unipotential

Base connections:

Pin No.	---	1	2	3	4	5	6	7	8	Cap
Element	---	sd & base sleeve	h	nc	nc	g	nc	h	k	a

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec	Pp	(A) Ehk	Rk	(A) Alt
Unit:	V	Vdc	Vdc	W	v	Ohms	ft
Maximum:	6.9	8,000	-200	12	300	---	(See note 1)
Minimum:	5.7	---	---	---	---	---	
TEST CONDITIONS:	6.3	8,000	0	---	300	9,600	---

GENERAL:

Qualification - Required

(A) denotes changes

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1							
1266	Total grid current	Eb = 6,000 V (see note 2)	0.65	II	Ic	0	-0.5	μAdc
1231	Emission	Eb = Ec = 30 Vdc (see note 2)	0.65	II	Is	32	---	mAdc
1256	Electrode current(cathode)	t = 60; Eb = 6,000 V (see note 3)	0.65	II	Ik	0.9	1.6	mAdc
(A)1201	Short and discontinuity detection		0.4	II	---	---	---	---
	Quality conformance inspection, part 2							
1301	Heater current		---	---	If	275	325	mA
1336	Heater-cathode leakage		---	---	Ihk	---	30	μAdc
1306	Transconductance	Eb = 1,500 Vdc; Ec = -1 Vdc; Rk = 0	---	---	Sm	675	1,125	μmhos
1316	Amplification factor	Eb = 1,500 Vdc; Ec = -1 Vdc; Rk = 0	---	---	Mu	380	520	---
1331	Direct-interelectrode capacitance	No shield	---	---	Cgp Cin Cout	0.60 4.0 1.2	0.90 6.0 2.6	pF pF pF
(A)1211	Insulation of electrodes		---	---	---	---	---	---
1031	Low-frequency vibration	Eb = 300 Vdc; Ec = 0; Rp = 10,000 ohms	---	---	Ep	---	200	mVac
(A)1111	Base pin solder depth		---	---	---	---	---	---
(A)1101	Secureness of base, cap, or insert		---	---	---	---	---	---
(A)1105	Permanence of marking		---	---	---	---	---	---
	Quality conformance inspection, part 3							
---	Life-test provisions	Group C; Ehk = 300 V	---	---	---	---	---	---
---	Life-test end point (500 hours)	Electrode current (cathode)	---	---	Ik	0.75	---	mAdc

NOTES:

1. See "Reduced pressure (altitude) rating," and altitude maximum peak voltage.
2. This test to be performed at the conclusion of the holding period.
3. Initial arcing may be tolerated but tubes shall be free of arcing at the conclusion of the test.

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Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - EL
Navy -
Air Force - 11, 85
DSA - ES

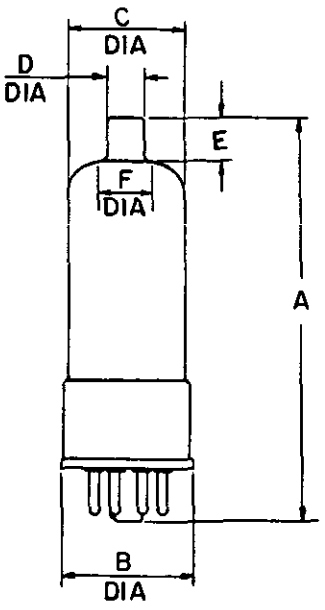
User activities:
Army - MI, WC
Navy - AS, OS, MC, CG, SH
Air Force - 19

Preparing activity: Navy - EC

Agent; DSA - ES

(Project 5960-2401-67)

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	3.500 (88.90)	3.875 (98.43)
B		1.312 (33.32)
Quality conformance inspection, part 3 (periodic-check)		
C	1.062 (26.97)	1.188 (30.18)
D	.355 (9.02)	.365 (9.27)
E	.406 (10.31) Nominal	
F	.420 (10.67) Nominal	

FIGURE 1. Outline drawing of electron tube type 2C53.

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